

Billing Compliance for Private Clinics

*A research prototype for detecting billing errors
in private healthcare — tested on real data*

Presented to Embryolab Fertility Clinic, Thessaloniki
Maria Polychroniadou · PhD Researcher · HealthLedgerAI
BSc International Management · MSc Accounting & Finance

THE PROBLEM

Private clinics lose money on billing errors they cannot see

Σφάλματα χρέωσης που περνούν απαρατήρητα

Every private clinic in Greece processes hundreds of invoices per month — to EOPYY, to private insurers, and directly to patients. Hidden inside that volume are systematic errors that go undetected because no one has time to check them manually.

→ **Duplicate invoices** — the same procedure billed twice to the same insurer, often months apart. Insurers eventually detect these and claw back payments with penalties.

→ **Upcoding** — a procedure billed above its allowed maximum rate under EOPYY or insurer tariffs. A common audit trigger that puts EOPYY contracts at risk.

→ **Phantom billing** — an invoice raised for a date when the patient has no recorded appointment. Can indicate data entry error or, in an audit, look like fraud.

→ **Unbundling** — procedure components that should be billed as one bundled code are billed individually at a higher combined total.

These are not hypothetical. In a dataset of 506 invoices modelled on private clinic billing data, the HealthLedgerAI prototype detected 70 alerts across 8 error types.

THE SOLUTION

HealthLedgerAI

Automated billing compliance detection — in seconds, not weeks

HealthLedgerAI is a Python-based prototype that reads a clinic's billing data (invoices, appointments, procedure rates, insurer networks) and automatically flags billing anomalies — before they reach an insurer, before an audit, and before a payment is made that will need to be reversed.

II

DETECTORS RUNNING
IN PARALLEL

506

INVOICES ANALYSED
IN THE TEST

70

ALERTS FLAGGED
AUTOMATICALLY

The output is a plain Excel report — no special software needed to read it. Every alert includes the invoice ID, the patient ID, what was found, and a recommended action. The clinic's billing team can work through it directly.

What it reads

Invoice records · Patient records ·
Appointment records · Procedure
rate tables · Bundle rules · Insurer
network lists

What it produces

Excel report with All Alerts, HIGH
Priority, and MEDIUM Priority tabs —
ready to hand to the billing team

What the output looks like

Examples from the prototype alert report — 506 invoices analysed

The following are real alerts generated by the prototype on a test dataset. Each finding is assigned a risk level and a recommended action.

D1 · DUPLICATE INVOICE — HIGH RISK

Vendor 'Radiology Dept' | Amount €145 | Date 2026-02-03 — billed twice to the same insurer on the same date.

Action: Verify with vendor. Reject duplicate before payment to avoid insurer clawback.

D2 · UPCODING — HIGH RISK

Procedure billed at €806 against a maximum allowed rate of €620. Excess of €186 per invoice.

Action: Withhold payment. Request clinical justification. Risk of EOPYY audit flag.

D4 · PHANTOM BILLING — HIGH RISK

Invoice raised for patient P047 on 2026-03-14, but no appointment record exists for that patient on that date.

Action: Verify patient was physically present. Do not submit to insurer without confirmation.

D11 · REPEATED SAME-DAY PROCEDURE — MEDIUM RISK

Same procedure billed 3 times on the same day for the same patient. May be legitimate — but needs clinical note confirmation.

Action: Verify clinical notes confirm multiple administrations before submitting to insurer.

All 70 alerts are available in the full Excel report. HIGH priority: 53 alerts. MEDIUM priority: 17 alerts.

Why fertility clinics are a natural fit

Γιατί οι κλινικές γονιμότητας έχουν ιδιαίτερα σύνθετη χρέωση

Fertility clinics operate in one of the most billing-intensive environments in private healthcare. A single IVF cycle generates multiple billing events across different categories — consultations, laboratory procedures, medications, anaesthesia, embryology services — each with different tariff rules.

→ **EOPYY + private insurers simultaneously** — IVF is covered by EOPYY up to 3 cycles for women under 50. This means clinics bill two different systems with different rules for the same patient. Mixed billing increases the risk of errors.

→ **High procedure values** — IVF cycle costs range from €3,000–€6,000. A single upcoding or duplicate billing error at this scale represents significant financial exposure.

→ **Multi-step billing sequences** — each cycle involves 8–12 separate billing events. Date mismatches and duplicate entries are common in high-volume, multi-step billing processes.

→ **ISO certification environment** — Embryolab is ISO 9001:2015 and EN 15224 certified, meaning it already operates to documented process standards. A billing compliance tool fits naturally into that framework.

WHAT I AM ASKING FOR

20 minutes and your honest opinion

Δεν πουλάω τίποτα — ζητώ εμπειρογνωμοσύνη

I am a doctoral researcher, not a salesperson. HealthLedgerAI is a research prototype I am developing as part of my PhD on billing compliance in private healthcare in Greece and Cyprus. I am here to learn from practitioners — not to sell a product.

I have signed a Non-Disclosure Agreement with you before showing this material, which means everything I share is protected and everything Embryolab shares with me stays within the boundaries of that agreement.

Specifically, I would like to ask:

1. Do the types of billing errors this prototype detects reflect real problems you encounter in your billing process?
2. Are there error types I have missed that are common in fertility clinic billing in Greece?
3. Would a tool like this — even at prototype stage — be useful to your billing or administration team?
4. Would you be willing to provide a short written statement that you reviewed this prototype as part of a research validation? (This is voluntary and for my doctoral research record only.)

Your feedback will directly shape the next version of this prototype. If the research progresses, I would welcome the opportunity to continue the conversation — but there is no obligation of any kind.

ABOUT THE RESEARCHER

Maria Polychroniadou

PhD Researcher · HealthLedgerAI Founder

Academic background

. BSc International Management

. MSc Accounting & Finance

. PhD candidate — AI & billing
compliance in private healthcare
(Greece & Cyprus)

. Erasmus for Young Entrepreneurs
participant — Thessaloniki, 2026

HealthLedgerAI

. Pre-launch research prototype

. 11 billing error detectors built and
tested

. Applied to EIT Jumpstarter 2026

. Target: EIC Pre-Accelerator 2027

healthledgerai.com · info@healthledgerai.com

All information in this presentation is covered by the NDA signed prior to this meeting.